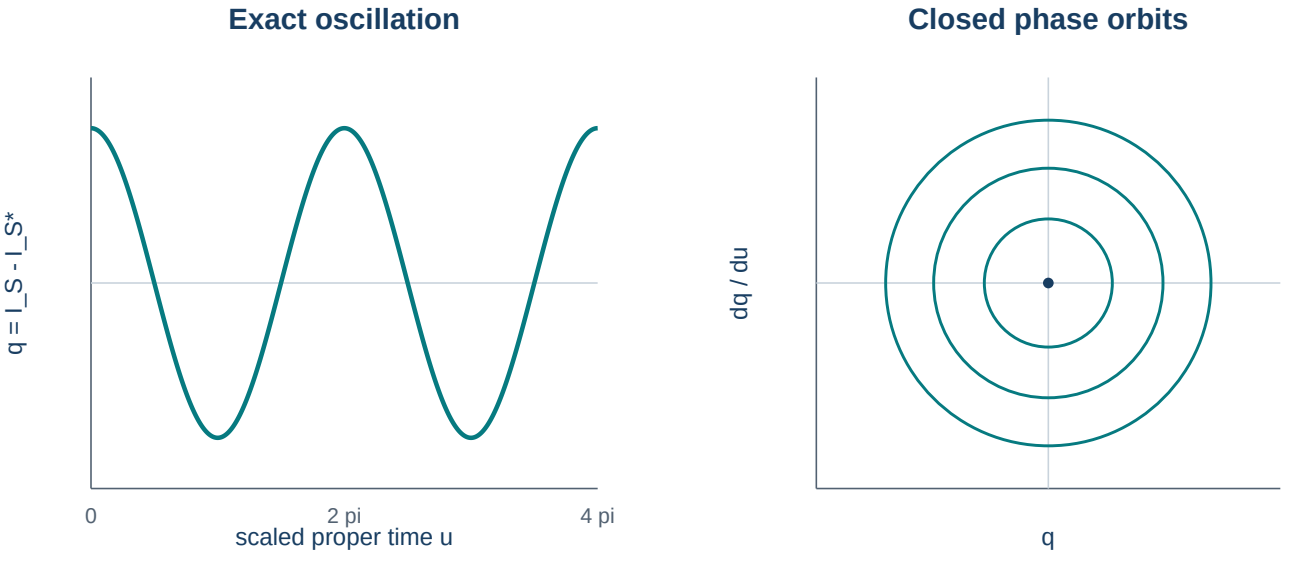

Analytic nondimensional illustration: q = cos(u). No empirical data or coupled-worldline solution.